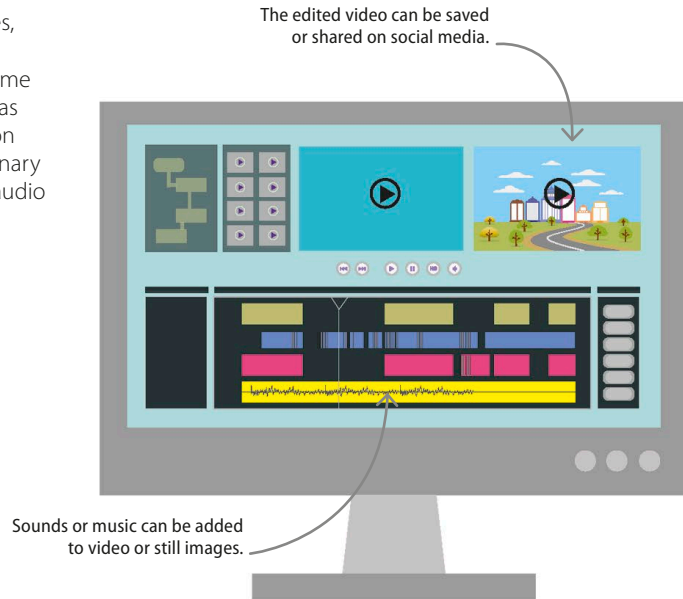


Encoding video files

A video is made from a series of still images, called frames, which are played one after another at a high speed. Frames are displayed at a constant rate, known as the frame rate – the equivalent to sample rate in an audio file. Just as with still images, a frame is made up of pixels, and data on the colour and brightness of each pixel is stored as a binary number. The number of bits used to store picture and audio information per second of video is called the bit rate.

▷ Post-production

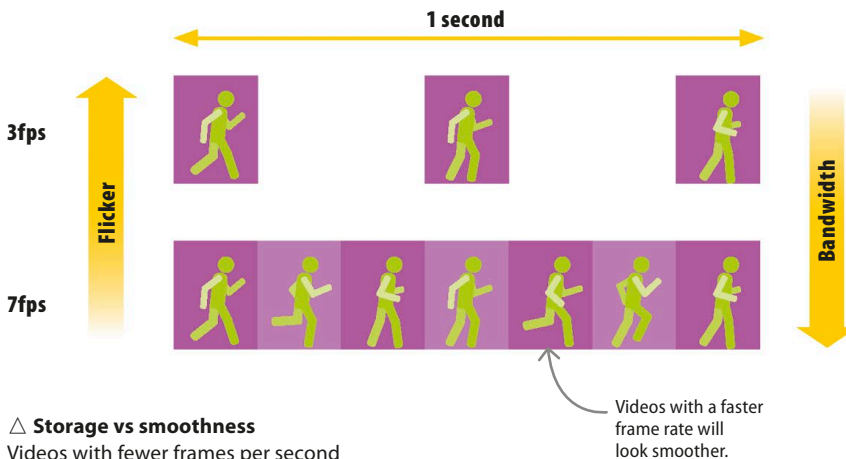
Video editing software allows people to import clips of video and audio files. These can be edited and arranged in any order. Users can add transitions between sections, and even include titles. Many programs also let users apply colour effects.



Frames per second

The frame rate of a video affects how smooth and realistic the motion appears to the human eye. The standard frame rate for film and digital video converted from analogue film is 24 frames per second (fps). Video created on digital cameras may be slightly faster, at 25 or 30 fps. People are so used to seeing 24 fps that higher rates can seem odd and unnatural.

Webcam **frame rates** are often **low** as this creates **less data** to transfer, saving **bandwidth**.



△ Storage vs smoothness

Videos with fewer frames per second require less data to store them, but the motion in these videos will look more jerky and unnatural.

IN DEPTH

Codecs

Encoding allows data to be stored or transferred, and decoding allows it to be played back. A program that can be used both to encode and decode digital data is called a codec. There are audio and video codecs, and each codec decodes from and encodes to a particular file format. H.265 and Xvid are examples of video codecs, while audio codecs include MP3 and AAC.